B.Sc. BOTANY — YEAR I

MAJOR COURSE III

APPLIED BOTANY

UNIT IV — Rural Development, Ethnobotany, Ethnomedicine, Ethnofibres and Ethno-food

Detailed Teacher-Style Notes

Definitions • Concepts • Tables • Flow diagrams • Examples • Applications • Exam points

Prepared in the same detailed format as the Major Course II notes.

UNIT IV — ETHNOBOTANY AND RURAL DEVELOPMENT

Syllabus topics covered: role of Botany in rural development; ethnobotany; ethnomedicine with Neem, Aloe, Clove, Ginger, Tulsi, Turmeric, Giloy, Amla, Ashwagandha and Arandi; ethnofibres with Areca, Coconut, Elephant grass and Cotton; ethno-food crops with Garadu, Singada, Kutki, Samak, Kodo, Bodo, Besan, Jowar, Makka, Bajra, Jau and Jai; fasting foods and ancient utensils.

4.1 Role of Botany in rural development

Definition — Rural development: A process of improving livelihoods, infrastructure, productivity, health, resource management and social well-being in rural communities.

Botany as a rural-development tool

Local plant resources → Scientific identification → Value addition → Cultivation / sustainable harvest → Income + nutrition + conservation

Botanical contribution	Rural-development outcome
Improved crop and nursery practices	Productivity and employment
Medicinal plant cultivation	Diversified income
Value addition to fibres and foods	Small enterprises
Agroforestry and restoration	Resource resilience
Plant biodiversity documentation	Conservation-linked livelihoods

4.2 Ethnobotany

Definition — Ethnobotany: The study of relationships between people, cultures and plants, including how plant knowledge is used, transmitted and adapted within communities.

A correct ethnobotanical record includes local name, botanical identity, family, plant part used, preparation method, cultural context, seasonality, source and conservation status. Traditional use should not be automatically presented as clinically proven medical treatment.

4.3 Ethnomedicine — syllabus examples

Plant	Botanical name	Family	Commonly recorded traditional-use theme
Neem	Azadirachta indica	Meliaceae	Hygiene, skin-related and other traditional preparations
Aloe	Aloe vera	Asphodelaceae	Leaf gel and topical traditional uses
Clove	Syzygium aromaticum	Myrtaceae	Aromatic spice and traditional dental/oral preparations
Ginger	Zingiber officinale	Zingiberaceae	Digestive and warming preparations
Tulsi	Ocimum tenuiflorum	Lamiaceae	Aromatic traditional household use
Turmeric	Curcuma longa	Zingiberaceae	Spice, colouring and traditional preparations
Giloy	Tinospora cordifolia	Menispermaceae	Traditional herbal preparations
Amla	Phyllanthus emblica	Phyllanthaceae	Food and traditional tonic preparations
Ashwagandha	Withania somnifera	Solanaceae	Traditional root-based preparations
Arandi	Ricinus communis	Euphorbiaceae	Oil-producing plant with traditional and industrial uses

4.4 Ethnofibres

Plant	Botanical name	Family	Fibre/use note
Areca	Areca catechu	Arecaceae	Leaf-sheath and other plant parts used in local products
Coconut	Cocos nucifera	Arecaceae	Coir fibre from mesocarp
Elephant grass	Cenchrus purpureus	Poaceae	Biomass/fibre potential and local utility
Cotton	Gossypium spp.	Malvaceae	Seed-hair fibre for textiles

4.5 Ethno-food crops

Local/common name	Botanical identity (where standard identification is clear)	Main importance
Garadu	Dioscorea spp. (regional name may vary)	Edible tuber; correct local species identification is necessary
Singada	Trapa natans complex / Trapa spp.	Aquatic edible fruit
Kutki	Regional name may vary; verify local species before scientific naming	Traditional/local grain use
Samak	Echinochloa frumentacea	Small millet used especially in fasting cuisine
Kodo	Paspalum scrobiculatum	Millet; drought-tolerant grain
Bodo/Besan	Food term rather than a single plant name; identify ingredient/source locally	Pulse-based food context
Jowar	Sorghum bicolor	Cereal grain
Makka	Zea mays	Cereal grain
Bajra	Cenchrus americanus	Pearl millet
Jau	Hordeum vulgare	Barley
Jai/Oats	Avena sativa	Cereal grain

4.6 Fasting foods and ancient utensils

For this activity, study foods actually used in the local community. Record ingredient, botanical source, processing method and perceived nutritional role. For ancient utensils, document material, plant/food use, maintenance and cultural context; do not claim health benefits without evidence.

Exam focus: Unit IV high-yield: ethnobotany definition and methodology; rural-development applications; local name + botanical name + family + use for all listed plants; distinguish traditional use from scientific clinical evidence.